

Borophene prefers to be flat because that's where its energy is lowest. It, however, needs the underlying substrate to make it wavy. When grown on a featureless surface, its natural form is flat like graphene. When grown on silver, borophene adopts its accordion-like form while silver reconstructs itself to match.

Once borophene is formed, it can easily be removed as the bond between silver and borophene is weak. This may allow atom-thick slices of borophene to be transferred to any substrate once it has been formed with the corrugations. This wavy conformation so far seems unique due to the exceptional structural flexibility and particular interactions of borophene with silver, and may be initially triggered by a slight compression in the layer when a bit too many boron atoms get onto the surface.

For the first time, research team has made composites and combined borophene with another material to create heterostructures as the fundamental building block for fabrication of electronic devices.

Applications

Borophene is both metallic and atomically thin, holding great promise for applications ranging from electronics to photovoltaics. 2D materials like graphene are touted as ideal for electronics but graphene being too flat and hard to stretch for flexible, wearable devices, wavy borophene might prove better than graphene. Borophene-based heterostructures will guide future and ongoing research into using borophene for functional nanoelectronic devices.

What the future holds

There is a need to develop reliable encapsulation and/or passivation schemes that allow borophene to be removed from an ultra-high-vacuum environment so that a practical device could actually be fabricated and tested. When borophene is on silver (both of which are metallic), the silver substrate electrically shorts borophene, which creates serious problems for any device application. This calls for transfer of borophene from the silver growth substrate to an electrically insulating substrate.

Borophene could be the key to the future wearable and stretchable devices, owing to its flexibility while exhibiting electrical properties. Materials that are not only flexible but can also stretch would enable electronics to be mounted onto any surface with much lower chances of breaking up. Superconductive materials are highly valuable in the wearable industry, where signals must flow without any issue. Therefore borophene may also find its way into the future of superconductors and, coupled with room-temperature superconductors, it may change the world forever. Hence more work is required before the full potential of borophene is realised in electronic applications. **EFY**

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